

HUJI Applicative Research Report — Agriculture & Food — October 2026

2 paper(s) · Agriculture & Food · Monthly · October 2026 · Generated October 01, 2026

■ ■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

Hebrew University researchers are addressing critical challenges in global agriculture, focusing on both increasing crop yields and optimizing resource efficiency. One promising project investigates the biological mechanisms behind premature fruit drop, aiming to develop novel interventions that could prevent substantial crop losses. Concurrently, another team is exploring advanced sensor technologies and AI to create precision irrigation systems that intelligently respond to plants' real-time water needs. These innovations hold significant commercial potential for enhancing food security and fostering more sustainable farming practices worldwide.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Alon Samach** — AgriTech, FoodTech, Synthetic Biology
- **Yotam Zait** — AgriTech, Software/AI

AGRICULTURE & FOOD — RESEARCH HIGHLIGHTS

1. Studying early events leading to fruitlet abscission.

12/50

"Breakthroughs in synthetic biology and plant genetics promise to halt premature fruit drop, significantly boosting crop yields."

Main researcher: Alon Samach

The Robert H. Smith Institute of Plant Sciences and Genetics in Agriculture, Faculty of Agriculture, Hebrew University of Jerusalem, Rehovot 7610001, Israel.

Published in Journal of experimental botany · 2026 Sep 16

ABOUT THIS RESEARCH

The research investigates the biological mechanisms and cellular processes that trigger the premature dropping of young fruit from plants. It focuses on understanding the early physiological triggers of fruitlet abscission to improve crop stability.

COMMERCIAL ANGLE

Developing chemical or genetic interventions to prevent premature fruit drop could significantly increase crop yields and stabilize food supply chains.

COMMERCIALISATION METRICS

Novelty

2/10

The title suggests a descriptive, fundamental biological study of a well-known physiological process rather than the introduction of a new technology, platform, or disruptive mechanism. Without an abstract, there is no evidence of a novel methodology or a groundbreaking paradigm shift in agricultural biotechnology.

Commercial Potential

3/10

The title suggests fundamental basic research into plant biology rather than a targeted technological solution. While understanding abscission could eventually inform crop yield optimization, there is no indication of a specific commercial application, IP strategy, or scalable product mentioned.

Market Size

3/10

The research focuses on a fundamental biological process in plant development (fruitlet abscission) which has niche industrial relevance in horticulture, but lacks evidence of a large-scale commercial application or high-value market driver.

Tech Readiness

2/10

The title suggests fundamental basic research into plant physiological processes rather than the development of a specific product or application. This work appears to be at the early discovery stage of understanding biological mechanisms.

IP Strength

2/10

The title suggests basic biological research into natural physiological processes, which is typically difficult to patent and lacks clear industrial application or platform defensibility.

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2. Stomatal aperture, VPD and transpiration - The chicken or the egg?

10/50

"AI-driven precision irrigation systems, informed by real-time plant data, are set to revolutionize water use efficiency in agriculture."

Main researcher: Yotam Zait · yotam.zait@mail.huji.ac.il.

The Robert H. Smith Institute of Plant Sciences and Genetics in Agriculture, Faculty of Agriculture, Food and Environment, The Hebrew University of Jerusalem, Rehovot 76100, Israel. Electronic address: yotam.zait@mail.huji.ac.il.

Published in Plant science : an international journal of experimental plant biology · 2026 Aug 31

ABOUT THIS RESEARCH

The research investigates the causal relationship between stomatal opening, vapor pressure deficit (VPD), and plant transpiration rates. It seeks to determine whether stomatal movement drives transpiration or if environmental humidity triggers the opening of these pores.

COMMERCIAL ANGLE

Developing precision sensor technologies or autonomous irrigation systems that can predict and respond to real-time plant gas exchange dynamics to optimize water use efficiency.

COMMERCIALISATION METRICS

Novelty

3/10

The title suggests a mechanistic study of established physiological feedback loops (VPD-stomatal regulation) rather than the introduction of a novel biological pathway or a disruptive technological platform. It appears to be a fundamental refinement of existing plant physiology knowledge rather than groundbreaking scientific innovation.

Commercial Potential

2/10

The title suggests a fundamental mechanistic study in plant physiology rather than a technology-driven application. Without evidence of a platform or tool, it appears to be basic academic research with limited direct commercial translation.

Market Size

2/10

The paper appears to be fundamental plant physiology research focused on basic biological mechanisms rather than a product or technology with a defined commercial application. There is no evidence of a scalable industrial or clinical market being addressed.

Tech Readiness

2/10

The title indicates a fundamental physiological study investigating causal relationships in plant biology, which represents basic scientific research rather than applied technology development. This work is at the earliest stage of knowledge generation (TRL 2) with no immediate path to a commercial product or industrial tool described.

IP Strength

1/10

The title suggests a basic physiological study of plant biology and environmental interactions, which describes natural phenomena that are inherently non-patentable. There is no indication of a novel tool, synthetic construct, or proprietary method that would provide industrial defensibility.

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